BEST KNOWN SQUARE PACKINGS

The square packing problem asks for the side $s(n)$ of the smallest square that can hold n unit squares, where the squares are free to rotate but cannot overlap

- **O** proved optimal (77 of 324)
- **=** exact value known (295 of 324)
- **≈** only known numerically (29 of 324)
- **R** rigid (22 of 324)

recent result, since August, 2026 (297 of 324)

90°	45°		
4	3	2	1	0

colors indicate distinct tilt angles

shade indicates number of full-side contacts

deg is the algebraic degree of that side length

Best packings due to Göbel, Trump, Wainwright, Hämäläinen, Stenlund, Friedman, Bidwell, Cantrell, Morandi, Ellsworth, Hajba, Schadt, Chaoweeraprasit, Couzo, Daniel, de Winter, Rehwaldt, Gupta, ry-xu, Chang, DeVincentis, hmbelvedere

Lower bounds due to Göbel, Stromquist, Kearney, Shiu, Bentz, Daniel, Vlasenko, Guzhou0806, Hosono, Ahmed, Karakuş, Ryu

Optimality proofs due to Göbel, Stromquist, Friedman, El Moumni, Kearney, Shiu, Bentz, Vlasenko, Levy, Daniel, Hosono, Ahmed, Karakuş; formalization contributions by Gurevitch, Guzhou0806; verification infrastructure and execution by EvolvingPrograms, Julian-JJ, Lewis

Diagram by Joshua Levy
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The Squares Project
github.com/jlevy/squares